EE3072 Dynamic Systems and Control

Midterm Exam Solutions

May 3, 2023

1. (25 pts) Determine the number of roots of the following polynomial that are in the right half-plane, in the left half-plane and on the $j\omega$ axis.

$$P(s) = s^6 - 2s^5 - 9s^4 + 6s^3 - 28s^2 + 56s + 96$$

Solution:

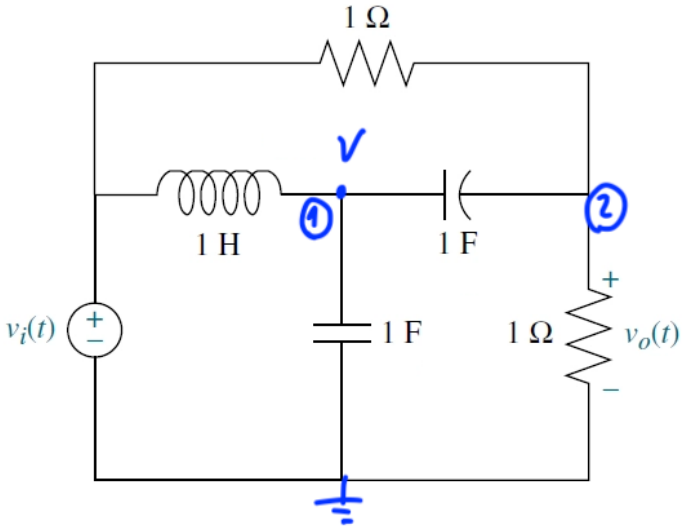
s^6	1	-9	-28	96
s^5	-2	6	56	
s^4	-6	0	96	
s^3	6	24		
s^2	24	96		
s^1	0 (48)			
s^0	96			

Table 1: Routh-Hurwitz Table

Since the s^1 row is completely zero, we need to take the derivative of the auxiliary polynomial and use its coefficients to proceed: $\frac{d}{ds}(24s^2 + 96) = 48s$.

- There are two sign changes in the first column: two poles are in RHP.
- Roots of the auxiliary polynomial are $24s^2 + 96 = 0 \implies s = \pm 2j$. Therefore two poles are on the imaginary axis.
- The remaining two poles are in LHP.

2. (25 pts) Consider the electrical circuit given in the following figure.



Obtain the transfer function of the system where v_i is the input and v_o is the output.

Solution:

From KCL at node 1, we have the following equation in s domain.

$$\frac{V_i(s) - V(s)}{s} = s(V(s) - V_o(s)) + sV(s)$$

From KCL at node 2, we have the following equation in s domain

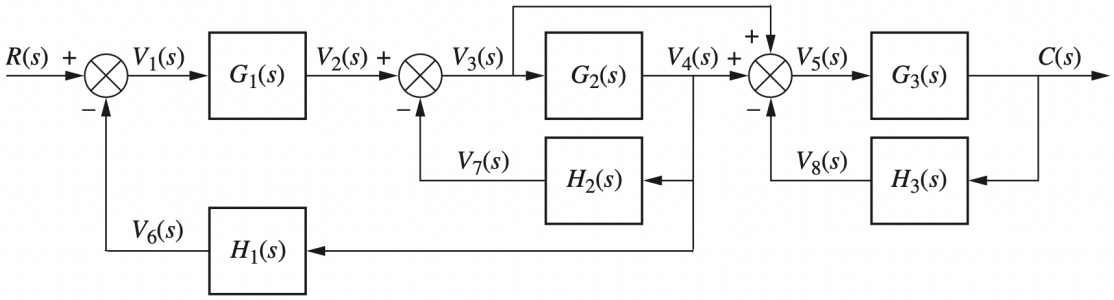
$$(V_i(s) - V_o(s)) + s(V(s) - V_o(s)) = V_o(s)$$

which implies $V(s) = \dfrac{(s+2)V_o(s) - V_i(s)}{s}$.

Substituting $V(s)$ into the first equation we obtain

$$\frac{V_o(s)}{V_i(s)} = G(s) = \frac{2s^2 + s + 1}{s^3 + 4s^2 + s + 2}.$$

3. (25 pts) Reduce the system shown in the figure below to a single transfer function.



Soluton:

There are two forward paths with the following gains:

FP1 $M_1 = G_1G_2G_3$

FP2 $M_2 = G_1G_3$

There are three loop paths with the following gains:

LP1 $l_1 = -G_2H_2$

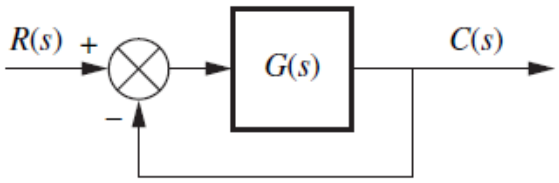
LP2 $l_2 = -G_1G_2H_1$

LP3 $l_3 = -G_3H_3$

and $\Delta = 1 - (l_1 + l_2 + l_3) + (l_1l_3 + l_2l_3)$. Furthermore, $\Delta_1 = \Delta_2 = 1$. From Mason's rule, the transfer function can be computed as

$$\begin{aligned} G(s) = \frac{1}{\Delta}\sum_{k=1}^{2} M_k\Delta_k &= \frac{1}{1 - (l_1 + l_2 + l_3) + (l_1l_3 + l_2l_3)}(M_1 + M_2) \\ &= \frac{G_1G_2G_3 + G_1G_3}{1 + G_3H_3 + G_2H_2 + G_1G_2H_1 + G_2G_3H_2H_3 + G_1G_2G_3H_1H_3} \end{aligned}$$

4. (25 pts) Sketch the root locus for the unity feedback system shown in the figure below where $G(s) = \dfrac{K(s+4)}{s(s+2)}$. Find the value of K such that the unit step response critically damped.



Solution:

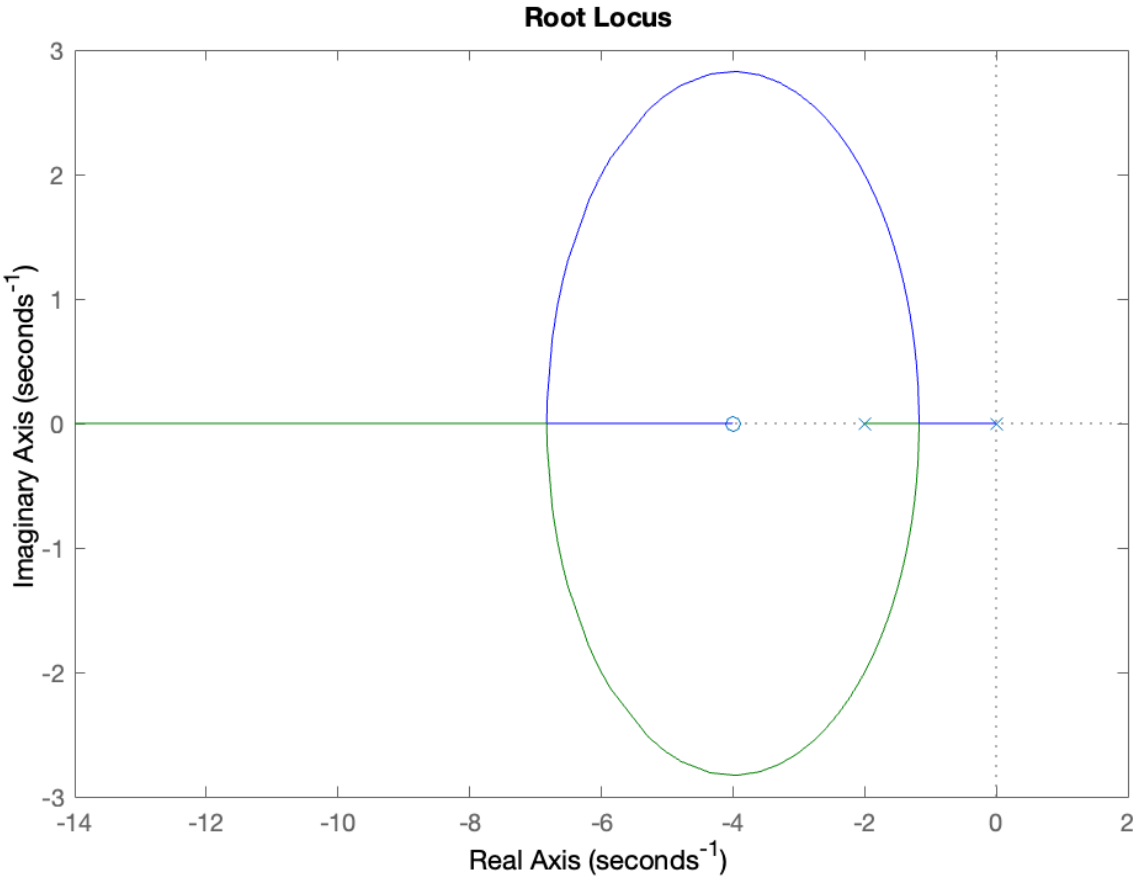
- The poles of loop transfer function: 0 and -2 which implies $n = 2$. The zero of the loop transfer function: -4 which implies $m = 1$. There will be two branches in the root locus.
- The points in the ranges $[0 - 2]$ and $(-\infty - 4)$ will be included in the root locus.
- Since $n - m = 1$, there will be one asymptote with an angle of 180°.
- The breakin and breakaway points can be found from the solution of

$$\frac{1}{\sigma - (-4)} = \frac{1}{\sigma - 0} + \frac{1}{\sigma - (-2)} \implies \sigma^2 + 8\sigma + 8 = 0$$

as $\sigma_1 = -4 - 2\sqrt{2}$ and $\sigma = -4 + 2\sqrt{2}$.

– Since there are no complex conjugate poles or zeros we do not need to compute the angles of arrival and departure.
– The root locus do not cross the imaginary axis and therefore we do not need to compute the critical value of K.

Root locus can be sketched as follows:



The unit step response will be critically damped when the closed loop poles are repeated roots, i.e., at the break-in and break-away points. The gain value at these points can be computed from the magnitude condition as follows: For $s = -4+2\sqrt{2}$, we have $K = \left.\frac{|s||s+2|}{|s+4|}\right|_{s=-4+2\sqrt{2}} = 0.34$. For $s = -4-2\sqrt{2}$, we have $K = \left.\frac{|s||s+2|}{|s+4|}\right|_{s=-4-2\sqrt{2}} = 0.66$.